Started on	2025-05-12 21:16
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Completed on	2025-05-12 23:56
Time taken	2 hours 39 mins
Grade	**10.00** out of 10.00 (**100**%)

Question **1**
Correct
Mark 2.00 out of 2.00

A business tax form is either filed on time or late, is either from a small or a large business, and is either accurate or inaccurate. There is an 11% probability that a form is from a small business and is accurate and on time. There is a 13% probability that a form is from a small business and is accurate but is late. There is a 15% probability that a form is from a small business and is on time. There is a 21% probability that a form is from a small business and is inaccurate and is late. There is a 12% probability that a form is inaccurate and is on time. There is a 17% probability that a form is from a large business and is accurate but is late. There is a 30% probability that a form is from a large business and is on time.

If a form is from a small business and is inaccurate, what is the probability that it was filed on time? (choose the closest value) 0.16 ✓

What is the probability that a form was filed late? 0.55 ✓

Question **2**
Correct
Mark 2.00 out of 2.00

A quality inspector selects a sample of 15 items at random from a collection of 100 items, of which 42 have excellent quality, 30 have good quality, 25 have poor quality, and
3 are defective.

What is the probability that the sample only contains items that have either excellent or poor quality? (choose the closest value) 136 ✓ $\times 10^{-5}$

What is the probability that the sample contains five items of excellent quality, five items of good quality, five items of poor quality, and no defective items?
254 ✓ $\times 10^{-4}$

Question 3

Correct

Mark 2.00 out of 2.00

A fair coin is tossed three times. A player wins $3 if the first toss is a head, but loses $1 if the first toss is a tail. Similarly, the player wins $1 if the second toss is a head, but loses $3 if the second toss is a tail, and wins (if head) or loses (if tail) $2 according to the result of the third toss. Let the random variable X be the total winnings after the three tosses (possibly a negative value if losses are incurred).

(a) Construct the probability mass function (put x_i in increasing order):

x_i	-6	-2	2	6		
p_i	1/8	3/8	3/8	1/8		

(b) The expectation is $E(X) =$ 0

(c) The variance is $Var(X) =$ 12

(d) The standard deviation is $SD(X) =$ 3.46

Question 4

Correct

Mark 2.00 out of 2.00

Five students are waiting to talk to the TA when office hours begin. The TA talks to the students one at a time, starting with the first student and ending with the fifth student, with no breaks between students. Suppose that the time taken by the TA to talk to a student has a normal distribution with a mean of 8 minutes and a standard deviation of 3 minutes, and suppose that the times taken by the students are independent of each other.

Suppose that the time that elapses between when the TA starts talking to the first student, and when the TA starts to have a headache, has a normal distribution with a mean of 15 minutes and a standard deviation of 5 minutes, which is independent of the times taken to talk to the students.

What is the probability that the TA's headache starts at a time after the TA has finished talking to the second student? 0.4404

What is the probability that the TA's headache starts at a time after the TA has finished talking to the third student? 0.1056

Question 5

Correct

Mark 2.00 out of 2.00

Suppose that $X \sim B(10, 1/3)$ is a random variable and consider the point estimate $\hat{p} = \frac{X}{11}$.

What is the bias of this point estimate? -1/33

What is the variance of this point estimate? 20/1089

Find the mean square error of this point estimate 21/1089